



Android devices have captured the smart products market and are to be found everywhere. Windows developers who wish to develop Android apps can use open source developer tools that work on the Windows platform. This article introduces some of the best among these.

This article briefly discusses tools that Android developers can use on the Windows operating system. There are six different categories of tools that facilitate Android app development using the Windows OS:

1. Source code management
2. Build tools
3. Application debugging tools
4. IDE
5. Continuous integration
6. Continuous code inspection

Source code management

A source control management system (SCM) provides version control for code that has been developed by the team, for the test scripts created by the testing team, for the configuration files created by the DevOps team, and so on. SCM provides collaboration, coordination and services between the different stakeholders involved in application life cycle management.

GitHub

GitHub is a hosted version control service that uses Git (visit <https://github.com/>).

Reasons for using GitHub: Git repositories are used for version control. The Git server helps to manage multiple repositories (when multiple people work on the same repository) to keep track of changes in code. GitHub also provides a facility for social code by providing the Web interface for the Git repository and management tools for collaboration.

Workflow of GitHub: Git uses the same centralised pattern as a subversion (SVN) workflow, but it is a more powerful development presenter than the latter. In Git, every developer has a local copy of the entire project, where each developer can work independent of all other changes made to the project. Commits can be added to the local branch and the main development/master branch can be ignored until it is convenient.

Initialising the central repository: The third stage in Figure 1 is the developer local branch.

Try to use the Git terminal, although the Git GUI application is available. I know the Git GUI application might be easier, but the terminal has proved to be more convenient in my experience as a developer. In the Git terminal, the command line serves to communicate with the user and gives a proper error message even when the command fails; and it also points out the proper solution.

Stable release	N/A – Hosted service
Owner	Microsoft
Website	https://github.com/
Written in	Ruby
On-premise or hosted	Git repository hosting service
Main features	Private and public repositories. GitHub supports source code, wikis, notifications, integration with automation tools, issue tracking, and so on.